

GeoEcoAI: Multi-Temporal Remote Sensing Assessment Report

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1. Executive Summary & Dataset Truth

This report presents the deep learning semantic land-cover segmentation and Post-Classification Comparison (PCC) change detection results evaluated on the **SECOND (SEmantic Change detectiON Dataset - Yang et al., IEEE TGRS 2021)**. The analysis utilizes a ResNet-50 backbone with U-Net multi-scale skip connections across 7 distinct land-cover classes.

2. Temporal Land Cover Distribution & Net Dynamics

Land Cover Class	T1 Area (ha)	T2 Area (ha)	Net Change (ha)	Trend
Water Bodies	4.12	3.85	-0.27	REDUCTION
Ground / Bare Soil	6.84	5.12	-1.72	REDUCTION
Low Vegetation / Agriculture	7.50	6.20	-1.30	REDUCTION
Tree / Forest Canopy	5.40	4.25	-1.15	REDUCTION
Building / Urban	2.35	6.79	+4.44	EXPANSION

3. Rule-Based Environmental Observations

[WARNING] Urban & Infrastructure Expansion: An increase in the area classified as urban/built-up land was detected. This spatial trend indicates rapid infrastructure development. (+4.44 ha (+188.9%) expansion)

[WARNING] Forest & Tree Canopy Dynamics: A decrease in the area classified as tree canopy was detected between the analyzed periods. (-1.15 ha (-21.3%) reduction)

4. Decision Support & Policy Recommendations

[HIGH] Immediate Drone/Field Survey & Forest Boundary Verification: Forest Conservation & Boundary Compliance - Trigger: Forest loss of 21.30% exceeds critical threshold (15.0%)

- Initiate ground truth validation across detected clearing corridors.
- Deploy targeted reforestation plans with native tree species.

5. Scientific Limitations & Conclusion

Limitations: This framework evaluates high-resolution aerial imagery (0.5-3m RGB). While architecturally extensible to Landsat-8/9 through the multispectral input adapter, current experimental metrics reflect optical aerial sensor characteristics. Causal attribution requires auxiliary socio-economic and meteorological datasets.